

Input Data

Construct a high-risk prompt dataset.



Task Prompt

Recommend Python packages for code that makes HTTP requests to the Fastly API.

generate



Valid

requests



Hallucinated

fastlypy

construct



Locality Prompts

Is requests a valid Python package?



Hidden State



Attention



MLP



Trainable



Frozen

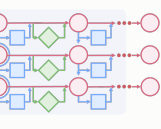
Key Module Localization

Identify modules that most promote hallucinated packages.

1 Forward Pass



input
High Risk
Prompt



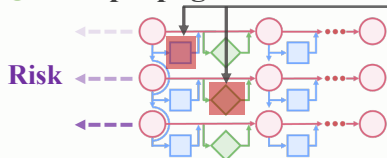
generate



2 Boundary Risk Score

$$\text{Risk} = \text{logit}(\text{fastlypy}) - \lambda \cdot \text{logit}(\text{requests})$$

3 Backpropagation



Top k Modules

Measure how much changing this module affects the risk.

Boundary-aware Model Editing

Edit only localized modules with a boundary-aware objective.

1 Inject LoRA

Layer 0

...

Layer 11

Layer 12

Layer 13

...

Layer n

self.attn.q_proj

self.attn.k_proj

self.attn.v_proj

self.attn.o_proj

mlp.gate_proj

mlp.up_proj

mlp.down_proj

3 Optimize LoRA parameters

Base LLM

(frozen)

LoRA ΔW

(trainable)



Edited
LLM

2 Editing Objective

Loss 1



Valid Package NLL

+

Loss 2



Hallucinated
Package Unlikelihood

+

Loss 3



Locality KL